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RESEARCH

AI-driven fuzzy requirement engineering for cybersecurity: integrating linguistic decision models

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Abstract
Cybercrime presents an escalating threat, necessitating innovative solutions beyond conventional security measures. Traditional approaches often fall short in addressing the complex and evolving nature of cyber threats. This paper proposes a novel methodology that integrates Artificial Intelligence (AI) and fuzzy logic for cybersecurity requirement engineering, specifically focused on modeling and prioritizing the requirements of the ChatGPT system. A fuzzy TOPSIS-based framework was developed, enabling stakeholders to express preferences using linguistic variables, which were then transformed into Triangular Fuzzy Numbers (TFNs) for structured evaluation. The functional and non-functional requirements were elicited using a goal-oriented strategy and modeled through use-case and class diagrams. The purpose of this study is to address the inherent ambiguity and subjectivity in stakeholder inputs by employing fuzzy logic, thus enhancing the accuracy and adaptability of requirement modeling in cybersecurity systems. A software application was also developed using VC + + to automate the ranking process. The findings demonstrate that the proposed fuzzy-based approach effectively captures stakeholder preferences and supports the prioritization of critical requirements. The use of fuzzy TOPSIS allows for clear differentiation among requirements based on multiple evaluation criteria. These findings are significant as they provide a practical and scalable solution for requirement modeling in cybersecurity applications, facilitating better decision-making, improving system resilience, and promoting proactive threat management in AI-integrated systems like ChatGPT.

Keywords Functional requirements · Non-functional requirements · Modeling · Goal-oriented modeling · Class diagram · UCD · Cybersecurity · ChatGPT

1 Introduction

Cybercrime is a relentless threat, continuously evolving and damaging individuals, businesses, and governments. Traditional security measures, often reactive and struggling to keep pace, necessitate innovative solutions [1]. Artificial intelligence (AI) presents a promising approach, offering proactive and automated methods for detecting and preventing cyberattacks. This paper delves into the potential of AI, specifically focusing on ChatGPT and fuzzy logic to bolster cybersecurity [2]. ChatGPT has evoked a mixed response within the scientific community and academia, reflecting the ongoing debate surrounding the benefits and risks of advanced AI technologies [3–8]. ChatGPT and similar language models can potentially improve conversational and writing tasks, resulting in enhanced output efficiency and accuracy. However, concerns have been raised regarding potential biases in ChatGPT's training data, which may limit its capabilities and lead to factual inaccuracies, including scientifically

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plausible yet misleading outputs (referred to as hallucination). Moreover, security concerns and the potential misuse of LLMs for spreading misinformation and cyber-attacks should be considered [9]. As observed from evolutionary and social psychology perspectives, the human tendency to resist change explains the immediate concerns and debates surrounding the widespread adoption of ChatGPT [10].

ChatGPT has attracted attention across various disciplines. In education, its release has sparked discussions on the potential obsolescence of essays as assignments [11]. In healthcare practice and academic writing, the focus has been on addressing factual inaccuracies, ethical issues, and the fear of misusing ChatGPT, including disseminating misinformation [12–14]. Integrating a fuzzy-based cybersecurity approach with ChatGPT for requirements modeling of a specific set of requirements holds both promise and challenges. In this context, the authors aim to explore the potential benefits and limitations of using fuzzy logic in ChatGPT for requirement modeling. This introduction overviews the authors' motivation, limitations, and contributions regarding applying a fuzzy-based cybersecurity approach in ChatGPT requirement modeling. The driving force behind this manuscript stems from the following factors:

(i) *Increasing Cybersecurity Threats*—With the ever-growing complexity and sophistication of cyber threats, there is a pressing need to enhance the capabilities of AI systems like ChatGPT in addressing cybersecurity requirements. Fuzzy logic offers the potential to handle uncertain and imprecise information, making it a suitable candidate for modeling requirements in this domain.
(ii) *Enhancing Requirement Understanding*—The authors recognize that understanding and capturing requirements accurately is crucial for effective cybersecurity. By leveraging fuzzy logic, ChatGPT can potentially interpret and analyze linguistic variables used by stakeholders, leading to a more nuanced understanding of specific requirements. This can facilitate improved requirement modeling and system development.
(iii) *Flexibility in Requirement Representation*—Fuzzy-based modeling allows for the representation of requirements that involve imprecise or ambiguous information. This flexibility can enable ChatGPT to capture and process requirements with inherent uncertainties or linguistic variability. The system can provide more adaptable and context-aware cybersecurity solutions by accommodating fuzzy requirements.

Based on the literature review, we have identified the following limitations.

(i) *Precision and Ambiguity Challenges*—Fuzzy logic introduces imprecision, which can potentially hinder the precision and clarity of requirement modeling. The authors acknowledge that striking a balance between accommodating imprecise information and maintaining the necessary level of precision is a significant challenge. Ensuring that fuzzy-based requirements are well-defined and can be interpreted consistently becomes critical.
(ii) *Formal Verification Complexity*—Traditional cybersecurity practices often rely on formal verification techniques to ensure compliance and security guarantees. Integrating fuzzy-based requirements with formal verification processes can introduce complexities due to fuzzy logic's inherent ambiguity and imprecision.

The authors recognize that addressing these challenges is essential to establishing confidence in the effectiveness of the proposed fuzzy-based approach. The authors contribute by investigating the application of fuzzy logic in ChatGPT for requirement modeling in the cybersecurity domain [15]. They delve into the specific challenges and opportunities this approach presents, aiming to shed light on its feasibility and effectiveness. Through an in-depth analysis, the authors identify the key issues that may arise when using ChatGPT for requirement modeling with a fuzzy-based cybersecurity approach. These issues include precision limitations, interpretation variability, formal verification complexities, system integration challenges, required expertise, scalability concerns, and the need for validation and assurance. By discussing the issues and challenges, the authors provide insights into the practical implications of using fuzzy logic in ChatGPT for cybersecurity requirement modeling. This information can guide developers, researchers, and organizations in making informed decisions regarding adopting and implementing a fuzzy-based approach.

The following are some of the practices of AI-based cybersecurity:

- Threat intelligence
- Behavioral analysis
- Predictive analytics
- Automation
- Cybersecurity training

ChatGPT Generated Strategies: ChatGPT generated the following strategies to combat cybercrime:

- Continuous monitoring
- Collaboration
- Hybrid AI
- Explainable AI
- Cyber Insurance

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Cybercrime is a growing problem affecting individuals, businesses, and governments worldwide. Traditional security measures are reactive and cannot keep up with the rapidly evolving threat landscape [16]. AI-based cybersecurity provides a proactive and automated approach to detecting and preventing cyberattacks. This research aims to explore the practices of AI-based cybersecurity and provide strategies generated by ChatGPT to combat cybercrime. The research is limited by data availability and the quality used to train AI algorithms. The effectiveness of AI-based cybersecurity also depends on the accuracy of threat intelligence and the ability to keep up with the evolving threat landscape. Additionally, the research is limited by the scope of ChatGPT's generated strategies and the potential challenges of implementing them in real-world scenarios [17, 18].

The research paper makes the following contributions:

- A methodology has been devised for the modeling of a sizable chosen collection of requirements specific to ChatGPT.
- A software application has been developed utilizing the Visual C++(VC++) programming language to determine the prioritization of ChatGPT requirements using the fuzzy TOPSIS technique.
- UCD and class diagrams have been utilized to model and analyze the structural and behavioral elements of a ChatGPT system.
- Highlighting the importance of continuous monitoring, collaboration, hybrid AI, explainable AI, and cyber insurance in improving the overall cybersecurity posture of organizations.
- Emphasizing the need for ongoing research and development to improve the effectiveness of AI-based cybersecurity in the face of evolving threats.

This paper is structured as follows. Section 2 delves into relevant existing research. Section 3 outlines the proposed approach for utilizing ChatGPT within the cybersecurity domain. Section 4 details the case study research methodology employed. Section 5 presents a comparative analysis of various language models. Section 6 discusses the potential of ChatGPT and AI for the future of cybersecurity. Finally, Sect. 7 concludes the paper by summarizing key findings, highlighting the significance of continuous research and development, and suggesting potential avenues for future exploration.

2 Related previous work

AI in cybersecurity has emerged as a promising approach to combat cybercrime. This literature review highlights some of the key research in AI-based cybersecurity practices and ChatGPT-generated strategies to combat cybercrime. Cybercrime is a growing threat, and traditional security measures are insufficient to counter it. AI is being used to detect and prevent cyberattacks. The following literature review highlights some of the key research in AI-based cybersecurity.

Sarker et al. (2021) [19] discussed the importance of continuous monitoring in AI-based cybersecurity. They highlight the need for real-time network traffic monitoring and user behavior to detect and prevent cyberattacks. The authors also emphasize the importance of collaboration between stakeholders, such as IT teams, security teams, and management, to improve the effectiveness of AI-based cybersecurity.

ChatGPT-generated strategies for cybersecurity have also been proposed. For example, in a study by Yoo et al. (2022) [20], ChatGPT was used to generate strategies for combating phishing attacks. The authors demonstrate that ChatGPT can generate effective phishing attack scenarios and provide actionable strategies to prevent and mitigate these attacks.

Bécue et al. (2021) [21] proposed a hybrid AI approach for cybersecurity. They combine machine learning algorithms with rule-based systems to improve the accuracy and efficiency of cybersecurity. The authors also highlight the importance of explainable AI, which enables cybersecurity experts to understand the reasoning behind AI-based decisions and improve the effectiveness of cybersecurity.

In another study by Jayathilaka et al. (2025) [22], ChatGPT was used to generate cybersecurity strategies for small and medium-sized enterprises (SMEs). The authors demonstrate that ChatGPT can provide customized cybersecurity strategies based on SMEs' unique characteristics and needs.

Ozkan Okey et al. (2024) [23] presented a comprehensive review of AI-based cybersecurity. They discuss the various applications of AI in cybersecurity, such as threat detection, network security, and malware detection. The authors highlight the benefits of AI-based cybersecurity, including improved accuracy, efficiency, and real-time response. However, they also point out the limitations of AI-based cybersecurity, such as the potential for false positives and the need for high-quality training data.

Sharma et al. (2022) [24] presented a survey of cybersecurity technologies based on AI. The authors discuss the different types of AI algorithms used in cybersecurity, such as supervised and unsupervised learning, deep learning, and reinforcement learning. They also discuss the various applications of AI in cybersecurity, such as intrusion detection, malware detection, and vulnerability assessment. The authors highlight the benefits of AI-based cybersecurity, such as improved accuracy and speed, but also note the limitations of AI algorithms, such as the potential for adversarial attacks and the need for explainability.

Mohamed. (2025) [25] focused on explainable AI for cybersecurity. The authors argue that explainability is crucial for building trust in AI-based cybersecurity. They discuss

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the challenges of explainability, such as the complexity of AI algorithms and the need to balance accuracy with interpretability. The authors highlight the importance of developing explainable AI techniques for cybersecurity, such as using visualization tools and developing interpretable models.

Lano and Rahimi [26] conducted a case study where they aimed to improve the design quality of class diagrams through refactoring. Various approaches have been developed to simplify the modeling of system requirements (SRs). One such approach, utilized by Narawita and Vidanage [27], involves using natural language processing techniques to generate UML diagrams from textual requirements automatically. This approach allows system analysts to save time by automating the process.

Mohammad et al. [28] developed a technique to extract information from use cases and convert it into class diagrams. The goal of this method is to identify different classes and their relationships. Similarly, Alsaadi [29] presented an approach for assessing software performance by utilizing Unified Modeling Language (UML) class diagrams. This approach involves predicting performance values using use cases and their associated data. Z notations, a notation for requirements modeling, have been employed to analyze systems in [30], facilitating the modeling of both static and dynamic perspectives. To expedite the development process, UML diagrams are often reused, as constructing them from scratch can be time-consuming. The concept of diagram similarity measure has been employed to identify reusable diagrams. A technique focused on evaluating the similarity between two diagrams by considering structural and lexical information was developed by Fauzan et al. [31]. Evaluation of structural similarity is based on the connections between actors and use cases.

Additionally, lexical information in UML diagrams has been measured using cosine similarity and other methods. A grammar-based strategy using a string comparison metric was discussed by Javed et al. [32] for validating class diagrams. This approach allowed for feedback and modification of class diagrams based on stakeholder suggestions. To enhance understandability, UML models should be kept simple, and multiple diagrams are often constructed at different levels of abstraction to ensure consistency. The consistency problem was solved by Faitelson et al. [33] by developing a set of guidelines known as diagram refinement. Through three case studies, Thakur and Gupta [34] showed the applicability of their tool-supported method for automatically creating sequence diagrams from natural language use cases. Karaa et al. [35] created a tool to extract class diagrams from natural language system requirements because extracting class diagrams from natural language requirements is a difficult operation.

Existing approaches in system requirements (SRs) modeling primarily focus on utilizing UML models and generating diagrams from textual requirements [35]. However, when faced with many requirements during software development, it becomes challenging to model and implement all of them due to organizational limitations [37]. This selection process involves multiple stakeholders expressing their preferences using linguistic variables. To tackle this issue, we propose a fuzzy-based method that enables modeling and implementing a chosen set of requirements. In our approach, TFNs represent linguistic variables for their simplicity [38]. It is worth noting that there is currently no literature available on modeling the selected set of ChatGPT requirements using UML diagrams within a fuzzy environment. Therefore, this paper aims to fill this gap by presenting an approach to select and prioritize ChatGPT requirements for modeling based on their high priority.

Table 1 presents a brief overview of the key findings from the reviewed studies, highlighting the success of ChatGPT in academic writing, healthcare education, and healthcare practice/research.

In summary, the literature highlights the importance of continuous monitoring, collaboration, hybrid AI, and explainable AI in improving the effectiveness of AI-based cybersecurity. ChatGPT-generated strategies have also shown promise in providing customized and actionable cybersecurity strategies. However, additional research is required to assess the efficacy of these approaches in practical contexts.

A cross-domain study of the use of ChatGPT is provided in Table 2, which demonstrates that it has potential in such areas as construction, education, healthcare, cybersecurity, and research. Although research has documented the positive aspects of using them, including better planning, personalized learning, pre-diagnosed cases, and even detection of threats, issues of reliability of data used, ethical application, academic dishonesty, and interpretability remain a concern. Some reviews also underline the importance of responsible deployment and research in the future to overcome the limitations that still exist.

3 Proposed approach for ChatGPT system

Figure 1 demonstrates the general scheme of work of the proposed requirements engineering framework. It starts with the involvement of several stakeholders (SH_1 to SH_n) in the process of requirements elicitation that is facilitated through a model focused on goals that is expressed as AND/OR decomposition. These invoke FRs, which are subsequently judged by decision-makers (D_1 to D_m) in a matrix form of input. It is on this basis that the requirement values (RVal) of each functional requirement are computed. Lastly, the FRs and

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Table 1 The main findings from the studies based on applications, limitations, and optimal solutions are summarized

Sources	Blueprint	Applications	Limitations	Optimal Solution
[39]	News explainer	Free bundle; well-structured content with respectable references	The traditional approach to educational assessment is nearing its conclusion, raising concerns about its impact on human knowledge and capabilities	Modifying educational assessments to give priority to critical thinking or reasoning skills
[40]	An Analysis of ChatGPT's Role in academic writing	ChatGPT aids in the literature review procedure by reducing time requirements, offering paper summarization capabilities, and improving language proficiency	Possible drawbacks of using ChatGPT include the potential for inaccurate information, the presence of bias, a potential decline in critical thinking and creativity within the scientific domain, ethical concerns, and the risk of plagiarism	ChatGPT's substantial contribution enables it to be acknowledged as an author
[41]	Concise evaluation of ChatGPT's suitability for academic writing in the field of biomedicine	Responses that are original, precise, and accurate, accompanied by a systematic approach; beneficial for enhancing topic clarity through training; time-efficient; fostering motivation for writing	Instances where instructions were not followed accurately; omission of proper in-text referencing and inclusion of inaccurate references; absence of practical examples and personal experiences; lack of depth in responses	ChatGPT has the potential to enhance academic writing skills; foster the principles of universal design for learning; appropriate usage of ChatGPT under academic guidance
[42]	Opinion piece discussing the advantages and drawbacks of ChatGPT	Enhance linguistic quality and readability	Excessive level of detail; presence of incorrect or biased information; potential for generating misleading errors; dissemination of misinformation; risk of plagiarism; concerns regarding the integrity of scientific records	The widespread adoption of ChatGPT is unavoidable; it is essential to accurately document its usage; ChatGPT should not be attributed or referenced as an author or co-author
[43]	Opinion piece discussing the significance of cybersecurity in safeguarding medical data	Flexibility; effectiveness; production of high-quality generated text; cost-effectiveness; potential for innovation; enhanced decision-making; improved diagnostic capabilities; predictive modeling; tailored healthcare approaches; streamlined clinical workflows for improved efficiency; remote monitoring	Data security issues	Highlighting the importance of cybersecurity in safeguarding medical data

their respective RVal scores will be represented in the UML tools or the use-case diagram and class diagram to illustrate the system structure and the interaction of stakeholders in a suitable manner.

The following list outlines the sequential actions of our method.

- **Step 1:** The process begins by recognizing the relevant parties involved and determining the objectives of ChatGPT.
- **Step 2:** The next step involves employing a goal-oriented approach to extract FRs and NFRs specific to ChatGPT.
- **Step 3:** Evaluation of FRs and NFRs for ChatGPT by DMs.

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Table 1 (continued)

Sources	Blueprint	Applications	Limitations	Optimal Solution
[44]	Opinion piece discussing the advantages and drawbacks of ChatGPT	Creating medical reports; offering a concise overview of medical records; composing correspondence for insurance providers; enhancing the comprehensibility of CAD systems	Hallucination (the generation of misleading yet scientifically plausible information); the necessity for meticulous formulation of questions or prompts; potential for inaccurate or incomplete outcomes; reliance on the training data; bias; potential for research misconduct	While recognizing the tremendous utility of ChatGPT, it is crucial to exercise caution in harnessing its capabilities
[45]	Information sharing regarding the utilization of ChatGPT in medical education and clinical decision-making	Automated assessment; aid in instruction; enhanced individualized learning; support in research endeavors; creation of clinical scenarios; quick information retrieval; language translation; documentation in clinical settings; assistance in clinical decision-making; personalized healthcare approaches	Absence of human-like comprehension; constrained knowledge restricted to the year 2021	While ChatGPT can provide valuable assistance in medical education, research, and clinical practice, it is important to acknowledge that it cannot substitute human capabilities
[46]	An all-encompassing editorial written prior to the release of ChatGPT, discussing the significance of theory-based research	Can assist in overcoming language obstacles; effectively handle large-scale data processing within tight timeframes; stimulate human creativity through the application of "AI in the Loop: Humans in Charge."	Ethical issues, legal accountability factors, lack of empathy and direct interaction, and inadequate transparency are areas of concern	The involvement of humans in the scientific process holds an inherent value that cannot be substituted by AI contributions
[47]	Viewpoint: The role of ChatGPT in translational research	Improved writing productivity, analysis of large datasets like electronic health records or genomic data, identification of factors influencing disease risk, and forecasting disease outcomes	The adequacy of available data; incapacity to comprehend the intricacies of the biological system	The integration of ChatGPT in scientific and medical journals is expected to grow in the coming years
[48]	A case study crafted with the aid of ChatGPT	Assisted in generating a concise and pertinent introductory overview	Inaccessibility to pertinent literature; restricted knowledge limited to 2021; potential inaccuracies in citations; constrained capacity to critically analyze results	At present, ChatGPT does not serve as a substitute for independent literature reviews in scientific research
[49]	Headline: ChatGPT's influence in the academic sphere	Beneficial for conducting literature reviews, analyzing data, and facilitating translation tasks	Ethical considerations, concerns regarding data privacy and security, potential bias, and transparency challenges	ChatGPT holds the promise of advancing academia; it is imperative to deliberate on the responsible and ethical utilization of ChatGPT

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Table 1 (continued)

Sources	Blueprint	Applications	Limitations	Optimal Solution
[50]	Opinion: Key areas of focus for ChatGPT research	Facilitated innovation; heightened efficiency in publishing timelines; potential for promoting equity in science; enhanced diversity of scientific viewpoints; greater availability of time for experimental design; optimization of academic training	Potential compromise of research quality; challenges regarding transparency; dissemination of misinformation; potential for inaccuracies, bias, and plagiarism; ethical considerations; potential for future monopolistic tendencies; absence of transparency	A prohibition on ChatGPT would be ineffective; establish regulations to ensure accountability, integrity, transparency, and honesty; thoughtfully evaluate the essential academic skills for researchers; expand the discourse within academia; an initiative is required to address the development and responsible utilization of LLMs in research
[51]	Opinion piece discussing the utilization of ChatGPT in scientific writing	ChatGPT assists in surmounting language barriers, fostering inclusivity in research endeavors	Ethical considerations regarding ghostwriting, questionable accuracy, and challenges with citations	Approach this innovation with an open mindset; authors should possess an adequate understanding of leveraging AI tools
[52]	WAME recommendations on ChatGPT	It can be a useful tool for researchers	Inaccurate or nonsensical responses; limited knowledge confined to pre-2021; absence of legal identity; potential for plagiarism	ChatGPT does not fulfil the ICMJE criteria and cannot be acknowledged as an author; authors should be transparent about their utilization of ChatGPT and assume responsibility for the content generated; editors require suitable detection tools to identify ChatGPT-generated content
[53]	Considering the influence of ChatGPT, there is a perspective regarding the future of medical writing	Increased efficiency in medical writing	Incomplete understanding of the medical field; ethical concerns; potential for prejudice; legal consequences; difficulties in ensuring transparency	A potent instrument in the medical realm; nonetheless, it is imperative to acknowledge its various limitations
[54]	An editorial piece exploring ChatGPT and the anticipated transformation of medical writing	Enhanced effectiveness in medical writing; language translation	Ethical concerns, plagiarism concerns, lack of originality, potential for misinformation, and bias risk	"AI in the Loop: Humans in Charge"

- **Step 4:** Fuzzy TOPSIS, a technique that uses fuzzy logic to ascertain the relative ranks of the requirements, is used to calculate the ranking values of the FRs.
- **Step 5:** The chosen set of FRs is modeled using use-case and class diagrams, which provide visual representations to depict the functionalities and relationships within the system.

3.1 Proposed algorithm for ChatGPT system

1. **Input:** DM, SH, Alternative, Criteria
2. **Output:** Ranking_Order
3. DM ← Select _Decision _Maker (SH)
4. NFReq ← Elicit _NFReq(Criteria);
5. FReq ← Elicit _FReq(Alternative);
6. EC ← Evaluate _Criteria (NFReq)
7. W ← Evaluate _Weights (EC)
8. EA ← Evaluate _Alternatives (FReq, NFReq, DM)
9. FDM ← Create _Fuzzy _Decision _Matrix (EA)
10. NFDM ← Normalized _Fuzzy _Decision _Matrix (FDM)
11. WNFDM ← Weighted _Normalized _Fuzzy _Decision Matrix (NFDM, W)
12. FPIS ← Calculate FPIS (WNFDM)
13. FNIS ← Calculate FNIS (WNFDM)
14. Ranking _Order ← Computations Ranking Order (FPIS, FNIS)
15. Return Ranking _Order

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Table 2 Comparative analysis based on key findings

Reference	Focus	Methodology	Key Findings
Saka et al. (2023) [55]	Construction industry	Case study	Opportunities in design, planning, and project management; limitations in data availability and cost; use case validation in quantity surveying
Rejeb et al. (2024) [56]	Education	Web mining and machine learning	Positive impact on personalized learning and content creation; challenges in information overload and potential for misuse
Caruccio et al. (2024) [57]	Medical diagnosis	Comparative study	ChatGPT shows promise in early diagnosis but lacks reasoning and clinical judgment compared to predictive models
Stahl & Eke (2024) [58]	Ethics	Literature review	Ethical concerns regarding bias, transparency, accountability, and potential for misuse
Habibi et al. (2023) [59]	Higher education	Survey	Positive student acceptance and perceived usefulness for learning activities
Peres et al. (2023) [60]	Research, teaching, and practice	Literature review	Potential impact on research methods, teaching approaches, and professional practice across various disciplines
Darkhabani et al. (2023) [61]	Autoimmunity research	Literature review	Potential for accelerating knowledge discovery and hypothesis generation
Albayati (2024) [62]	Student perceptions	Survey	Students perceive ChatGPT as a helpful tool but express concerns about the potential for academic dishonesty
Okey et al. (2023) [63]	Cybersecurity	Topic modeling and sentiment analysis	Potential for threat detection and analysis, but limitations in accuracy and interpretability
Tayan et al. (2023) [64]	Higher education technology courses	Critical review	There is a need to adapt courses to consider the capabilities and limitations of AI language models
Ali et al. (2023) [65]	Knowledge and innovation	Scoping review	Knowledge gaps and challenges related to the responsible development and deployment of ChatGPT
Menon & Shilpa (2023) [66]	User adoption	UTAUT model analysis	User intention to use ChatGPT is influenced by performance expectancy, effort expectancy, social influence, facilitating conditions, and hedonic motivation
Scanlon et al. (2023) [67]	Digital forensics	Literature review	Potential benefits in automating tasks and identifying relevant data but concerns about reliability and potential for manipulation
Sohail et al. (2023) [68]	Taxonomy and future directions	Literature review	Categorization of existing research, identification of key challenges (bias, explainability, safety), and suggestions for future research directions
Ray (2023) [69]	Comprehensive review	Literature review	Overview of ChatGPT's background, applications, challenges, limitations, and future scope
Khennouche et al. (2024) [70]	Deploying ChatGPT and chatbots	Literature review	Insights and challenges in deploying ChatGPT for FAQs and customer service chatbots

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The proposed algorithm for the ChatGPT system involves selecting a decision maker and eliciting functional and non-functional requirements based on the provided criteria. Evaluation criteria and weights are determined, and the alternatives are assessed using a fuzzy-based approach. The ranking order of the alternatives is computed by calculating the fuzzy positive and negative ideal solutions. The algorithm provides a concise and systematic method for ranking alternatives in the ChatGPT system based on given inputs and evaluation processes.

Step 1: The process begins by recognizing the relevant parties involved and determining the objectives of ChatGPT.

This stage aims to recognize various individuals or organizations involved in a project and their respective goals. Stakeholders are individuals or entities who possess a vested interest in the project's advancement. The stakeholders have been categorized for this study based on their relative levels of interest (Intr) and influence (Inf) on the project. These categories include (i) stakeholders with high influence and high interest, (ii) stakeholders with high influence but low interest, (iii) stakeholders with high interest but low influence, and (iv) stakeholders with low influence and low interest. The set of stakeholders participating in eliciting FRs and NFRs for ChatGPT is denoted as SH = {SH1, SH2,…, SHn}.

Step 2: The next step involves employing a goal-oriented approach to extract FRs and NFRs specific to ChatGPT

The objective-driven approach has been utilized to elicit functional requirements (FRs) and non-functional requirements (NFRs) for ChatGpt (G). This approach involves breaking down and improving the main objective by decomposing it into sub-goals (SGs) and assigning responsibilities to specific agents or systems. These SGs are interconnected using "logical AND" and "logical OR" connectives, forming an AND/OR graph. When SG "t" is linked to multiple SGs through logical AND connectives, all connected SGs must be fulfilled for SG "t" to be achieved. Conversely, with a logical OR connective, the accomplishment of any SG can lead to the attainment of SG "t" [71].

Step 3: Assessment of FRs and NFRs for ChatGPT by DMs

Let p = {DM_1, DM_2, …, DMp} represent the set of DMs responsible for evaluating the elicited FRs and NFRs for ChatGPT. In practical applications, these DMs often utilize linguistic variables to assess the FRs and NFRs for ChatGPT [72–74]. To represent linguistic variables, many types of fuzzy numbers, including TFNs, trapezoidal fuzzy numbers, sigmoidal fuzzy numbers, pentagonal fuzzy numbers, and others, are utilized. In our study, we have specifically opted for TFNs to model linguistic variables throughout the computational process. We have chosen TFNs due to their simplicity, enabling easy understanding and representation during computation and analysis.

This choice was made because TFNs have a unique advantage in modeling and interpreting data simultaneously during computation in the early stages of modeling. Consider other fuzzy representations like trapezoidal or pentagonal fuzzy numbers; while they provide more detail, they also increase the difficulty of computation and the level of detail required to combine information. Prior studies (e.g., [75]) have indicated that TFNs preserve the stability of aggregation and in the decision-making processes, which makes them appropriate for prioritization of requirements in the context of cybersecurity. As a result, TFNs were used to facilitate the decision-making processes in cybersecurity requirements prioritization while preserving the depth of analysis. As an alternative to justify the use of TFNs, a simulation-based comparative evaluation of other fuzzy models is suggested as further research.

Step 4: Fuzzy TOPSIS, a technique that uses fuzzy logic to ascertain the relative ranks of the requirements, is used to calculate the ranking values of the FRs

The FRs obtained, denoted as FReq1, FReq2, …, FRrqx, are assessed based on the NFRs, represented as NFReq1, NFReq2, …, NFReqy, by a group of stakeholders (SH). This evaluation aims to construct a "fuzzy decision matrix" (FDMtx). Each decision maker, denoted as SHz where z ranges from 1 to Z, expresses their fuzzy rating using TFNs, which can be represented as [T]_z = (i_z,j_z,k_z) for z = 1, 2, …, Z, with the membership function μ_T (x). Subsequently, the fuzzy ratings are aggregated, resulting in a calculated value denoted as T (i, j, k) for z = 1, 2, …, Z. Here,

$$i = \min\{i_z\},\ j = \frac{1}{z}\sum_{z=1}^{Z} j_z,\ \text{and}\ k = \max\{k_z\} \tag{1}$$

Suppose the fuzzy rating of the zth SH for a FR is $R_{nmp} = (i_{abz}, j_{abz}, k_{abz})$, where $a = 1, 2, \ldots, x$, and $b = 1, 2, \ldots, y$. Here, x and y are the number of FRs and NFRs, respectively. The aggregated fuzzy rating (R_{ab}) of FRs concerning NFRs can be calculated as $(R_{ab}) = (i_{ab}, j_{ab}, k_{ab})$. Here,

$$i_{ab} = \min\{i_{abz}, \},\ j_{ab} = \frac{1}{z}\sum_{z=1}^{Z} j_{abz},\ k_{ab} = \max\{k_{abz}\} \tag{2}$$

where,

- iab: Represents the lowest bound of the aggregated TFN and is taken as the minimum of all lower bounds iabz across all stakeholders.
- jab: Represents the middle value of the aggregated TFN and is calculated as the average of the middle values jabz from all stakeholders.

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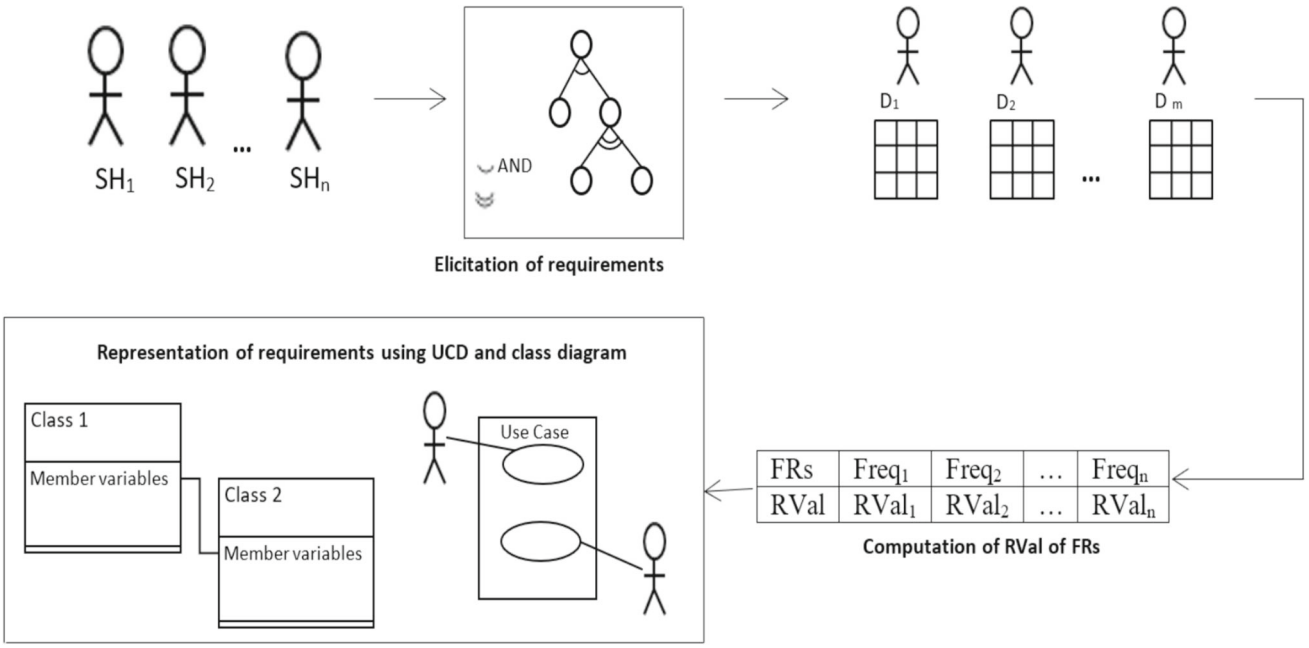


Fig. 1 Block diagram of the proposed approach

- kab: Represents the upper bound of the aggregated TFN and is taken as the maximum of all upper bounds kabz.

The importance weight of a specific NFR assigned by the zth SH represented as $wk_{abz} = (k_{bz1}, k_{bz2}, k_{bz3})$, is used to calculate the fuzzy weights (wk_{ab}) for each NFR:

$$wk_b = (wk_{b1}, wk_{b2}, wk_{b3}) \tag{3}$$

where,

$$wk_{b1} = \min\{i_{bz1}\},\ wk_{b2} = \frac{1}{z}\sum_{z=1}^{Z} j_{bz2},\ wk_{b3} = \max\{k_{bz3}\}. \tag{4}$$

The $FDMtx$ is constructed as:

$$FDMtx = \begin{bmatrix} f_{11} & f_{12} & \ldots & f_{1p} \\ f_{21} & f_{22} & \ldots & f_{2p} \\ . & . & & . \\ . & . & \ldots & . \\ . & . & \ldots & . \\ f_{q1} & f_{q2} & & f_{qp} \end{bmatrix}$$

$$WK = \left[wk, wk_2, \ldots, wk_y\right]$$

Here, $R_{ab} = (i_{ab}, j_{ab}, k_{ab})$ and $wk_b = (wk_{b1}, wk_{b2}, wk_{b3})$; $a = 1, 2, \ldots, x$; $b = 1, 2, \ldots, y$ can be approached by TFNs.

The normalized fuzzy decision matrix (normalized FDmtx) has been created by employing a linear scale transformation to normalize the FDmtx, i.e., $NormFDMtx$.

$$NormFDMtx = [h_{ab}]_{x\times y} a = 1,2, \ldots, xandb = 1,2, \ldots, y \tag{5}$$

where,

$$h_{ab} = \left\{\frac{i_{ab}}{k_b^*}, \frac{j_{ab}}{k_b^*}, \frac{k_{ab}}{k_b^*}\right\}$$

$$k_b^* = maxk_{ab}$$

The weights assigned to the NFRs are multiplied by the normalized fuzzy decision matrix (NormFDMtx) to generate the weighted normalized fuzzy decision matrix (G). This process can be defined as follows.

$$G = [g_{ab}]_{x\times y},\ a = 1,2, \ldots, xandb = 1,2, \ldots, y \tag{6}$$

$$g_{ab} = h_{ab}(.)wk_b$$

Here, wk_b Denotes the significance or priority assigned to each NFR through the importance weight. Using the weighted normalized fuzzy decision matrix (G), the values of the "fuzzy positive ideal solutions" (FPIDSoln) and "fuzzy negative ideal solutions" (FNIDSoln) are computed.

$$FPISoln = (FPISoln_1, FPISoln_2, \ldots, FPISoln_z) \tag{7}$$

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$$FNISoln = (FNISoln_1, FNISoln_2, \ldots, FNISoln_z) \tag{8}$$

where

$$FPISoln_b = \max\{FPISoln_{ab3}\}$$

$$FNISoln_b = \min\{FNISoln_{ab1}\}$$

$a = 1,2, \ldots, x$ and $b = 1,2, \ldots, y$

By utilizing the values of FPIDSoln and FNIDSoln, the closeness coefficient (CCoeff) values are computed. The ranking of the FRs is then determined based on these CCoeff values. The calculation of CCoeff can be expressed as follows.

$$CCoeff_a = \frac{d_a^{FNISoln}}{d_a^{FPISoln} + d_a^{FNISoln}}, a = 1,2, \ldots, x \tag{9}$$

where,

$$d_a^{FPISoln} = \sum_{b=1}^{y} d(g_{ab}, FPISoln_b) \quad a = 1, 2, \ldots, x \tag{10}$$

$$d_a^{FNISoln} = \sum_{b=1}^{y} d(g_{ab}, FNISol_b) a = 1,2, \ldots, x \tag{11}$$

The distance measurement (d) quantifies the separation or dissimilarity between two fuzzy numbers.

Step 5: The chosen set of FRs is modeled by utilizing use-case and class diagrams, which provide visual representations to depict the functionalities and relationships within the system

This step's objective is to rank the FRs according to the priority values that have been assigned to them. Use-case diagrams (UCDs) and class diagrams will be made using the FRs with the highest-ranking values for modelling purposes.

4 The process of conducting case study research

The process of conducting case study research (PCCSR) adheres to the recommendations put forth by Runeson et al. [76]. This process comprises three sub-processes: (a) design and planning of the case study, (b) preparation and collection of data, and (c) analysis of the gathered data. Numerous studies have employed these guidelines to carry out the PCCSR effectively. For instance, Sadiq [75] applied the guidelines proposed by Runeson et al. [76] to identify and analyze stakeholders in a fuzzy environment. The subsequent sections provide a detailed explanation of each sub-process within the PCCSR.

4.1 The phase of case study design and planning involves creating a framework and strategy for conducting the case study

The objective of the "case study design and planning" (CSDP) stage is to offer a comprehensive explanation and comprehension of the essential components involved, which include: (4.1.1) the objective or aim of the case study, (4.1.2) the case or subject under investigation, and (4.1.3) the research questions (RQs) that steer the study.

The linguistic variables presented in Figs. 2 and 3 have been used to generate fuzzy evaluation and weighting in the process of prioritizing requirements. The triangular membership functions of five linguistic terms, EL, L, A, H, and EH, are plotted on a numerical scale that measures 2 to 10, as in Fig. 2, to enable the evaluation of both FRs and NFRs. Figure 3, however, is a visualization of the same linguistic terms on a scaled normalization of 0 to 1, which is, in this case, specifically applied to the weighting values identifying NFRs. These two figures illustrate the way of making the qualitative judgments of stakeholders to be processed using fuzzy numerical values.

4.1.1 *Objective:* This study aims to create a model for a particular collection of ChatGPT's FRs and NFRs.

4.1.2 The case: The second aspect of CSDP involves establishing the scope or subject matter of the study. This study focuses on investigating ChatGPT, aiming to enable the implementation of various activities associated with ChatGPT. These activities include: (a) knowledge retrieval, (b) context awareness, and (c) Natural Language Understanding.

4.1.2 *RQs:* As a component of the CSDP sub-process, the following RQs have been formulated.

- **RQ 1:** Which set of requirements for ChatGPT should be considered when modeling and analyzing the FRs and NFRs of ChatGPT?
- **RQ 2:** Do both the UCDs and class diagrams effectively capture the same information regarding the FRs and NFRs of ChatGPT during the requirements analysis phase?

4.2 Preparation and collection of data

To gather DM perspectives for further study, FRs are evaluated in connection with NFRs. This study collected data by assessing the FRs with the input of five DMs, considering various NFRs. A detailed account of the data preparation and collection process is presented in the following section.

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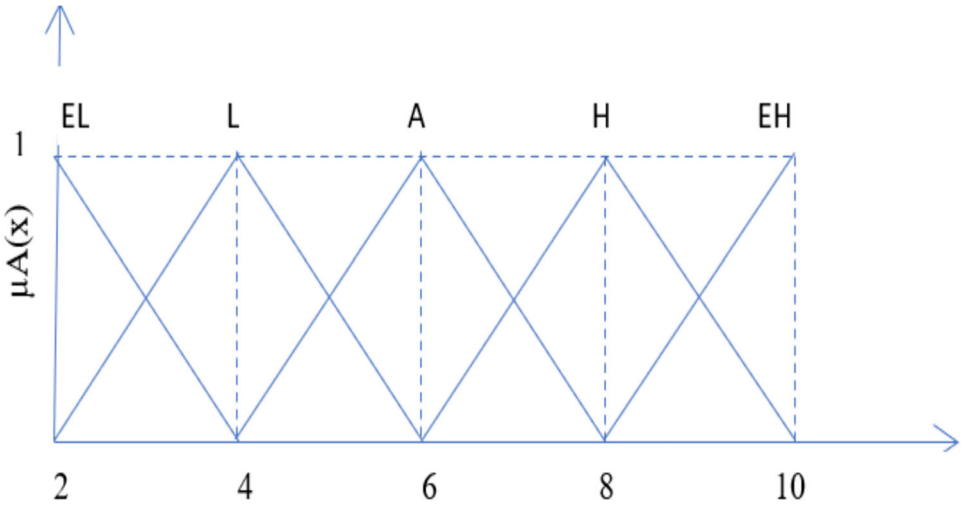


Fig. 2 Linguistic variables are employed for the evaluation of both the FRs and NFRs

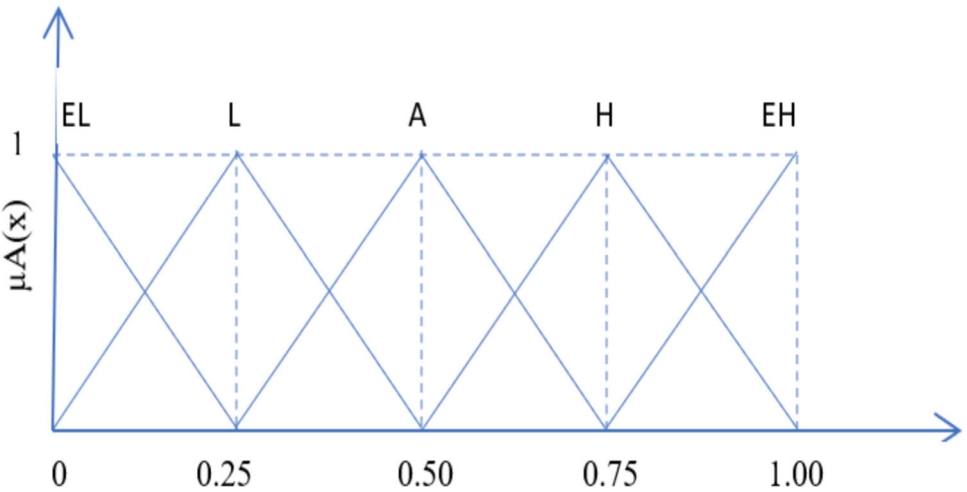


Fig. 3 A linguistic variable is employed to obtain the weight values for the NFRs

4.3 Data analysis process

Data collection encompassed the assessment of both FRs and NFRs with the participation of DMs to facilitate analysis. The collected data were then utilized to analyze the formulated RQs.

4.3.1 Step 1: The process begins by recognising the relevant parties involved and determining the objectives of ChatGPT

In the proposed work, eliciting requirements for a specific goal of ChatGPT involves identifying three stakeholders: SH1, SH2, and SH3. One of the stakeholders, SH1, consists of a team of requirements analysts responsible for gathering the FRs and NFRs of the ChatGPT system. On the other hand, SH2 represents the developer with considerable influence (Inf) and a keen interest (Intr) in the development process. The primary objective of SH1 is to automate the tasks performed by end users.

4.3.2 Step 2: The next step involves employing a goal-oriented approach to extract FRs and NFRs specific to ChatGPT

The goal of this step is to gather requirements based on stakeholders' demands. The "goal-oriented strategy," which attempts to elicit both the FRs and NFRs of the ChatGPT system, has been selected for this assignment. Breaking down and clarifying sub-goals using an AND/OR graph makes it easier to identify FRs and NFRs. The NFRs focus on quality criteria that characterize the non-behavioral components of the system, whereas the FRs pertain to ChatGPT's functionality. The FRs and NFRs of the ChatGPT system have been effectively found using the goal-oriented methodology. These specifications are presented in the provided list.

4.4 List of the FRs and NFRs:

$FReq_1$: Natural Language Understanding; $FReq_2$: Natural Language Generation; $FReq_3$: Context Awareness;

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Table 3 Linguistic variables are used to assign importance weights to each criterion

EL	0.00, 0.00, 0.25
L	0.00, 0.25, 0.50
A	0.25, 0.50, 0.75
H	0.50, 0.75, 1.00
EH	0.75, 1.00, 1.00

Table 5 Linguistic Variables for Rating

EL	(2.00, 2.00, 4.00)
L	(2.00, 4.00, 6.00)
A	(4.00, 6.00, 8.00)
H	(6.00, 8.00, 10.00)
EH	(8.00, 10.00, 10.00)

$FReq_4$: Multi-turn Conversation Support; $FReq_5$: Intent Recognition; $FReq_6$: Entity Recognition; $FReq_7$: Sentiment Analysis; $FReq_8$: Knowledge Retrieval; $FReq_9$: Error Handling; $FReq_{10}$: Personalization; $NFReq_1$: Performance; $NFReq_2$: Scalability; $NFReq_3$: Robustness; $NFReq_4$: Security; $NFReq_5$: Accuracy; $NFReq_6$: Ethical Use; $NFReq_7$: Extensibility; $NFReq_8$: Adaptability; $NFReq_9$: Usability; $NFReq_{10}$: Documentation.

4.4.1 Step 3: Evaluation of FRs and NFRs for ChatGPT by decision-makers

Five decision makers (DMs) were enlisted to evaluate the FRs and NFRs in this study using the linguistic variables "Extremely Low" (EL), "Low" (L), "Average" (A), "High" (H), and "Extremely High" (EH). For the evaluation procedure, TFNs represent these linguistic variables, as shown in Fig. 2. The TFN for the variable W, for instance, is (2, 4, 6). Ten stakeholders are involved in the evaluation of NFRs, using language characteristics such as "Extremely High" (EH), "High" (H), "Average," "Low," and "Extremely Low" (EL). Figure 3 shows the TFNs related to these linguistic characteristics. For instance, the TFN for the variable VH is (0, 0, 0.25). The language variable used to elicit the weights of NFRs is shown in Fig. 3. After analyzing the NFRs, the opinions of the ten stakeholders are gathered, allowing for the calculation of NFR weights. In addition, we acquired information from [77] to analyze the FRs.

To process conflicting inputs of different stakeholders and to aggregate them fairly, we have used the fuzzy aggregation techniques with the TFNs. These TFNs helped in normalizing a variety of linguistic expressions to create a possibility of the structured combination of varied points of view. The last stage of prioritizing was by weighted averaging of the stakeholders, where each view was given equal weight, thus avoiding bias from opinions. This process facilitates a consensus-based mechanism that is dynamic and iterative and is reflective of the negotiations involving requirements conducted in the real world.

Table 3 represents the linguistic variables used to assign importance weights to each criterion. The linguistic variables EL, L, A, H, and EH correspond to different levels of importance.

The values within each linguistic variable indicate the importance range assigned to a criterion, with 0.00 representing the lowest importance and 1.00 representing the highest importance. The three values within each linguistic variable represent the assigned importance range's lower, middle, and upper bounds.

Table 4 represents the importance of the criteria provided by 10 stakeholders for the corresponding NFReqs. The linguistic variables EL, L, A, H, and EH denote the criteria, representing different levels of importance. Each cell in the

Table 4 The importance weight of the criteria from 10 Stakeholders

NFReq1	EH	H	EH	H	A	L	A	EL	H	L
NFReq2	H	L	EH	EH	H	A	L	A	A	EL
NFReq3	A	EH	L	H	EH	H	EH	L	EL	A
NFReq4	A	EH	L	H	EH	H	EH	L	EL	A
NFReq5	A	EH	L	H	EH	H	EH	L	EL	A
NFReq6	A	EH	L	H	EH	H	EH	L	EL	A
NFReq7	A	EH	L	H	EH	H	EH	L	EL	A
NFReq8	A	EH	L	H	EH	H	EH	L	EL	A
NFReq9	A	EH	L	H	EH	H	EH	L	EL	A
NFReq10	A	EH	L	H	EH	H	EH	L	EL	A

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table indicates the linguistic variable assigned by a stakeholder for the importance of the corresponding criterion about the specific NFReq. The values assigned range from the lowest importance (EL) to the highest importance (EH), with intermediate levels represented by L, A, and H.

Table 5 represents the linguistic variables used for rating, indicating the range of values assigned to each variable. The linguistic variables EL, L, A, H, and EH correspond to different rating levels. The three values within each variable represent the assigned rating range's lower, middle, and upper bounds.

Table 6 presents the ratings provided by decision-makers (D1 to D5) for 10 alternatives, which consist of both non-functional requirements (NFReqs) and functional requirements (FReqs), across 10 criteria. The ratings are represented using linguistic variables (EL, L, A, H, EH) that indicate the level of preference or satisfaction assigned by each decision-maker for each alternative. The table provides an overview of how the decision-makers rated the different requirements under the various criteria, allowing for a comparison and analysis of their preferences and priorities.

4.4.2 Step 4.3.4: Fuzzy TOPSIS is utilized to calculate the ranking values of the FRs, employing fuzzy logic to determine their relative rankings

Using the fuzzy TOPSIS method, the functional requirements (FRs) ranking values were determined by considering the data presented in Tables 3, 4, 5, and 6. The fuzzy TOPSIS method has been implemented in the VC + + programming language to facilitate the ranking order analysis of datasets, regardless of their size. Firstly, we construct the FDMat by applying Eq. (1) to the ten FRs. Next, we calculate the weights of the NFRs using Eq. (2). Then, we obtain the NormFDMat by applying Eq. (4).

Table 7 represents the weighted normalized fuzzy decision matrix for 10 alternatives under different NFReqs and FReqs. Each alternative is assigned a weighted normalized fuzzy value, which is a combination of the normalized fuzzy values and the corresponding weights of the criteria. The values within the fuzzy sets indicate the lower, middle, and upper bounds of the respective weighted normalized fuzzy ranges. The table provides a concise representation of the weighted normalized fuzzy values for the alternatives in relation to the specific requirements considered.

Table 8 shows the distance between each Freq and the FPIDSol solution based on each NFReq. The distances are represented by numeric values. All the distances between the Freqs and FPIDSol are the same and equal to 0.78. This indicates that the Freqs have a consistent distance from the FPIDSol solution across all the NFReqs considered.

Table 9 shows the distances between each FReq and the FNIDSol solution based on each NFReq. The distances are represented by numeric values. All the distances between the FReqs and FNIDSol are the same and equal to 0.07 or 0.08. This indicates that the FReqs have consistent distances from the FNIDSol solution across all the NFReqs considered, but the specific distances vary between 0.07 and 0.08.

Table 10 presents the distances between each FReq and the optimal solution (Ai, A*) with respect to each NFReq. The distances are represented by numeric values. The table shows the specific distances between each FReq and the optimal solution for each NFReq. The distances vary across different FReqs and NFReqs, ranging from 0.34 to 0.57. These distances quantify the dissimilarity between each FReq and the optimal solution for each criterion.

Table 11 illustrates the distances between each FReq and the negative solution (Ai, A-) concerning each NFReq. The distances are measured using numeric values. The table displays the distances between each FReq and the negative solution for each NFReq. The distances range from 0.19 to 0.51 and differ across various FReqs and NFReqs. These distances represent the dissimilarity between each FReq and the negative solution for each criterion.

Table 12 provides close coefficients and ranking order of the FReqs. The closeness coefficient represents the degree of closeness of each FReq to the ideal solution. The ranking order indicates the relative position of each FReq based on its closeness coefficient. The FReqs are numbered from 1 to 10, and their corresponding closeness coefficients range from 0.42 to 0.59. The ranking order shows that FReq 8 has the highest coefficient and is ranked first, while FReq 6 has the lowest coefficient and is ranked last.

4.4.3 Step 4.3.5: The chosen set of FRs is modeled by utilizing use-case and class diagrams, which provide visual representations to depict the functionalities and relationships within the system

This section aims to utilize UCDs and class diagrams to model the FRs and engage in a discussion regarding the formulated RQs.

RQ 1: Which set of requirements for ChatGPT should be considered when modeling and analyzing the FRs and NFRs of ChatGPT?

Common requirements typically considered include language understanding, natural language processing, response generation, context awareness, security, scalability, performance, and user experience. However, it is essential to involve relevant stakeholders and conduct a thorough analysis to determine the specific requirements that are relevant and critical for the ChatGPT system being developed. In the development process, specific top requirements are chosen for modeling and analysis based on their ranking order. The stakeholders involved in the development process determine

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Table 6 Rating of the 10 alternatives by decision-makers under 10 criteria

NFReq's	FReq's	D1	D2	D3	D4	D5
NFReq1	FReq1	H	A	H	H	A
	FReq2	EH	L	EL	L	EL
	FReq3	EH	EL	L	EL	L
	FReq4	A	H	A	A	H
	FReq5	H	EH	H	H	H
	FReq6	L	EL	A	EL	L
	FReq7	H	EH	H	H	L
	FReq8	EH	H	EH	EH	H
	FReq9	H	H	H	H	H
	FReq10	H	EH	H	EH	EH
NFReq2	FReq1	H	H	H	H	EH
	FReq2	EH	H	EH	EH	EH
	FReq3	L	EL	EL	L	EL
	FReq4	H	H	A	H	A
	FReq5	L	EL	L	EL	EL
	FReq6	A	H	H	A	A
	FReq7	H	L	L	EL	L
	FReq8	EH	EH	H	EH	EH
	FReq9	H	H	H	EH	H
	FReq10	A	H	EH	EH	H
NFReq3	FReq1	H	L	A	L	A
	FReq2	EH	H	H	H	EH
	FReq3	EH	H	L	EL	L
	FReq4	A	H	H	A	H
	FReq5	L	EL	EL	L	EL
	FReq6	A	H	A	A	A
	FReq7	H	L	A	EL	A
	FReq8	EH	H	EH	EH	EH
	FReq9	H	EH	H	EH	H
	FReq10	H	H	H	EH	H
NFReq4	FReq1	H	EH	H	H	H
	FReq2	EH	H	H	EH	EH
	FReq3	EH	H	H	EH	EH
	FReq4	A	H	H	A	H
	FReq5	A	H	H	H	A
	FReq6	L	EL	EL	L	EL
	FReq7	H	A	H	A	A
	FReq8	EH	H	EH	EH	EH
	FReq9	EH	A	EH	EH	H
	FReq10	L	EL	L	L	L
NFReq5	FReq1	H	A	H	H	A
	FReq2	EH	H	H	H	H
	FReq3	EH	H	H	EH	EH
	FReq4	H	A	H	A	H
	FReq5	L	A	EL	L	EL

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Table 6 (continued)

NFReq's	FReq's	D1	D2	D3	D4	D5
	FReq6	A	H	H	H	A
	FReq7	H	L	EL	A	L
	FReq8	EH	H	EH	EH	EH
	FReq9	H	A	H	EH	H
	FReq10	H	L	L	A	H
NFReq6	FReq1	H	H	EH	EH	H
	FReq2	EH	H	EH	EH	EH
	FReq3	EH	EH	H	H	EH
	FReq4	L	A	EL	A	L
	FReq5	A	A	EL	H	A
	FReq6	H	H	L	A	H
	FReq7	H	H	H	L	A
	FReq8	EH	H	EH	EH	EH
	FReq9	A	H	H	A	H
	FReq10	L	A	EL	L	L
NFReq7	FReq1	H	A	H	H	H
	FReq2	EH	H	EH	EH	EH
	FReq3	EH	H	H	EH	EH
	FReq4	L	A	A	A	H
	FReq5	A	H	H	H	A
	FReq6	L	A	L	A	L
	FReq7	A	L	EL	L	EL
	FReq8	EH	H	EH	EH	H
	FReq9	H	EH	EH	EH	EH
	FReq10	L	A	A	L	A
NFReq8	FReq1	H	H	H	H	EH
	FReq2	EH	H	EH	EH	EH
	FReq3	H	H	H	EH	H
	FReq4	L	H	A	H	A
	FReq5	A	H	EL	A	H
	FReq6	L	EL	L	EL	EL
	FReq7	A	H	H	A	L
	FReq8	EH	H	EH	EH	EH
	FReq9	H	H	H	EH	H
	FReq10	L	A	A	H	A
NFReq9	FReq1	H	H	H	H	EH
	FReq2	EH	H	EH	EH	EH
	FReq3	EH	H	H	EH	EH
	FReq4	H	EH	H	H	H
	FReq5	L	A	H	A	H
	FReq6	A	L	EL	L	A
	FReq7	EH	H	EH	H	EH
	FReq8	EH	EH	EH	EH	H
	FReq9	H	H	H	H	A
	FReq10	L	A	A	L	A

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Table 6 (continued)

NFReq's	FReq's	D1	D2	D3	D4	D5
NFReq10	FReq1	H	L	H	H	A
	FReq2	H	H	H	H	EH
	FReq3	H	H	H	EH	EH
	FReq4	L	EL	L	L	EL
	FReq5	EL	L	L	EL	A
	FReq6	A	EL	L	A	L
	FReq7	H	A	H	L	L
	FReq8	EH	EH	EH	EH	H
	FReq9	H	EH	EH	H	H
	FReq10	L	A	A	A	L

the value of "n," which represents the number of requirements to be selected.

RQ 2: Do the UCDs and class diagrams effectively capture the same information regarding the FRs and NFRs of ChatGPT during the requirements analysis phase? Both UCDs and class diagrams can effectively capture information about ChatGPT's functional requirements (FRs) and non-functional requirements (NFRs) during the requirements analysis phase. However, they may capture different aspects of the requirements. UCDs primarily focus on representing user interactions, tasks, and goals, providing insights into the user experience and system usability. They emphasize the human-centric perspective and help understand how users interact with the system [78].

On the other hand, class diagrams depict the system's static structure, including classes, relationships, attributes, and methods. They primarily capture the system's internal structure, data models, and the relationships between various components. While UCDs provide insights into user interactions and goals, class diagrams offer a more detailed view of the system's structural components. By utilizing both UCDs and class diagrams, a comprehensive understanding of the FRs and NFRs can be obtained, covering both user-centric and system-centric aspects of the requirements. Figure 4 shows the proposed method using UCD. The class diagram of the proposed approach is shown in Fig. 5.

The functional and structural design of the proposed ChatGPT-based system is represented in Figs. 4 and 5, respectively. The Use Case Diagram (UCD) is depicted in Fig. 4 and demonstrates the interaction between the different types of users (new, web, and registered users) and the system functionalities, including the registration, logging in, authentication through the OTP, the handling of the questions, feedback submittals, and accounts. It also shows the writers of the system, such as ChatGPT, and the database. The Class Diagram laid out in Fig. 5 represents the internal hierarchy by defining major classes of the system: Registration, Questions, Answers, and Review and Feedback. Each of the classes includes appropriate attributes and methods, showing how user information, queries, responses, and feedback are navigated as a part of the system architecture. When combined, the figures give an all-inclusive user interaction and back-end logic.

5 Comparative analysis of different language models

Table 13 provides a comparative analysis of various language models (ChatGPT, GPT-3, BERT, Transformer-XL, and Bard), including their year of release, number of parameters, training data, pre-training method, and tasks/applications.

5.1 Comparative benchmarking with traditional methods

We carried out a comparative benchmarking study using AHP on the same set of FRs and NFRs to analyze the effectiveness of the proposed fuzzy TOPSIS-based method. AHP is one of the most popular multi-criteria decision-making techniques, which creates a weighting system through pairwise comparisons, yet it lacks in terms of vague reasoning or uncertainty. The results of the comparison are shown in Table 14, which depicts the difference in ranks and contested awarded reasons of the fuzzy TOPSIS method.

5.2 Cybersecurity best practices that can be implemented with the help of ChatGPT

ChatGPT has the potential to become a beneficial component of improving many cybersecurity best practices through one of these traits: the natural processing of language, contextual thinking, and synthesizing massive amounts of data.

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Table 7 Weighted Normalized Fuzzy decision matrix

NFReq1	FReq1	(0.14,0.42,0.78)
	FReq2	(0.07,0.26,0.78)
	FReq3	(0.07,0.26,0.78)
	FReq4	(0.14,0.39,0.78)
	FReq5	(0.21,0.49,0.78)
	FReq6	(0.07,0.21,0.62)
	FReq7	(0.07,0.44,0.78)
	FReq8	(0.21,0.53,0.78)
	FReq9	(0.21,0.46,0.78)
	FReq10	(0.21,0.53,0.78)
NFReq2	FReq1	(0.20,0.46,0.75)
	FReq2	(0.20,0.53,0.75)
	FReq3	(0.07,0.15,0.45)
	FReq4	(0.13,0.40,0.75)
	FReq5	(0.07,0.15,0.45)
	FReq6	(0.13,0.37,0.75)
	FReq7	(0.07,0.24,0.75)
	FReq8	(0.20,0.53,0.75)
	FReq9	(0.20,0.46,0.75)
	FReq10	(0.13,0.46,0.75)
NFReq3	FReq1	(0.08,0.34,0.78)
	FReq2	(0.23,0.53,0.78)
	FReq3	(0.08,0.34,0.78)
	FReq4	(0.15,0.43,0.78)
	FReq5	(0.08,0.17,0.47)
	FReq6	(0.15,0.38,0.78)
	FReq7	(0.08,0.31,0.78)
	FReq8	(0.23,0.58,0.78)
	FReq9	(0.23,0.53,0.78)
	FReq10	(0.23,0.50,0.78)
NFReq4	FReq1	(0.21,0.49,0.78)
	FReq2	(0.21,0.53,0.78)
	FReq3	(0.21,0.53,0.78)
	FReq4	(0.14,0.42,0.78)
	FReq5	(0.14,0.42,0.78)
	FReq6	(0.07,0.16,0.47)
	FReq7	(0.14,0.39,0.78)
	FReq8	(0.21,0.56,0.78)
	FReq9	(0.14,0.51,0.78)
	FReq10	(0.07,0.21,0.47)
NFReq5	FReq1	(0.13,0.40,0.75)
	FReq2	(0.20,0.46,0.75)
	FReq3	(0.20,0.51,0.75)
	FReq4	(0.13,0.40,0.75)
	FReq5	(0.07,0.20,0.60)
	FReq6	(0.13,0.40,0.75)
	FReq7	(0.07,0.26,0.75)
	FReq8	(0.20,0.53,0.75)
	FReq9	(0.13,0.44,0.75)
	FReq10	(0.07,0.33,0.75)
NFReq6	FReq1	(0.23,0.53,0.78)
	FReq2	(0.23,0.58,0.78)
	FReq3	(0.23,0.55,0.78)
	FReq4	(0.08,0.26,0.62)
	FReq5	(0.08,0.34,0.78)
	FReq6	(0.08,0.41,0.78)
	FReq7	(0.08,0.41,0.78)
	FReq8	(0.23,0.58,0.78)
	FReq9	(0.15,0.43,0.78)
	FReq10	(0.08,0.24,0.62)
NFReq7	FReq1	(0.15,0.46,0.78)
	FReq2	(0.23,0.58,0.78)
	FReq3	(0.23,0.55,0.78)
	FReq4	(0.08,0.36,0.78)
	FReq5	(0.15,0.43,0.78)
	FReq6	(0.08,0.29,0.62)
	FReq7	(0.08,0.22,0.62)
	FReq8	(0.23,0.55,0.78)
	FReq9	(0.23,0.58,0.78)
	FReq10	(0.08,0.31,0.62)
NFReq8	FReq1	(0.20,0.46,0.75)
	FReq2	(0.20,0.53,0.75)
	FReq3	(0.20,0.46,0.75)
	FReq4	(0.07,0.35,0.75)
	FReq5	(0.07,0.33,0.75)
	FReq6	(0.07,0.15,0.45)
	FReq7	(0.07,0.35,0.75)
	FReq8	(0.20,0.53,0.75)
	FReq9	(0.20,0.46,0.75)
	FReq10	(0.07,0.33,0.75)
NFReq9	FReq1	(0.23,0.50,0.78)
	FReq2	(0.23,0.58,0.78)
	FReq3	(0.23,0.55,0.78)
	FReq4	(0.23,0.50,0.78)
	FReq5	(0.08,0.38,0.78)
	FReq6	(0.08,0.26,0.62)

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Table 7 (continued)

	FReq7	(0.23,0.55,0.78)
	FReq8	(0.23,0.58,0.78)
	FReq9	(0.15,0.46,0.78)
	FReq10	(0.08,0.31,0.62)
NFReq10	FReq1	(0.08,0.41,0.78)
	FReq2	(0.23,0.50,0.78)
	FReq3	(0.23,0.53,0.78)
	FReq4	(0.08,0.19,0.47)
	FReq5	(0.08,0.22,0.62)
	FReq6	(0.08,0.26,0.62)
	FReq7	(0.08,0.36,0.78)
	FReq8	(0.23,0.58,0.78)
	FReq9	(0.23,0.53,0.78)
	FReq10	(0.08,0.31,0.62)

In the case of threat intelligence, ChatGPT may be helpful in summarizing and compiling threat feeds, research articles, and dark web chat logs into a list of actionable threat intelligence to identify new points of vulnerability or attack scenarios quickly. It can produce specific learning modules, customized tests, and simulations of the attacks to make employees understand and act in response to cyber threats. In the planning of incident response, ChatGPT may be utilized to write and edit more general response playbooks, making the actions very clear, uniform, and applicable to different threats. ChatGPT can be useful in network security as an administrator to create configuration checklists, log firewall and IDS/IPS rule modifications, and inform about best practices regarding network segmentation and access control. In the case of security testing, the model can be of use when it comes to suggesting test cases, induction of typical mistakes of configurations, and creating the scripts or payload concepts (under boundaries of ethics and law) that should be applied to penetration testing or hoax phishing mail to help suggest the resilience of the organization without causing any harm. Also, it may be used to develop the payload ideas of penetration tests, as it is possible to propose ways of emulating the safe yet realistic exploits taking place in the controlled environment. ChatGPT can create highly formatted incident response reports to report timelines, technical investigative details, and corrective actions appropriately to both the technical team and executive stakeholders. Lastly, in the case of security alerts, as they are presented in natural language, ChatGPT would turn raw and technical lists of events into summaries that non-technical decision-makers can use to appreciate the importance of an event and its magnitude. Through incorporating ChatGPT into such cybersecurity processes, organizations will be able to automate information processing, better conduct training, reach readiness, and, in the end, improve the overall organization's security.

6 Future of ChatGPT and AI

Elon Musk provided a strong, and some could argue exciting or terrifying, comment on the future of AI research when discussing how quickly AI has advanced, "Impressive progress has been made from the first GPT to the third GPT. If this pace of advancement is maintained, GPT-5 or 6 may become equal to the smartest humans". Recall that this statement was made in 2020. At the time of writing (2023), GPT-4 is anticipated to be published shortly. Rapid movement, change, and evolution are the norm.

AI's talent increased by 10,000% in the last three years, which should serve as a wake-up signal to everyone. Administrators, trustees, board members, teachers, professors, and everyone involved in higher education must recognize that this is a new reality, not a passing trend or something that will disappear. There cannot be any longing for it to disappear or for everything to end. This must be addressed head-on, and we must learn to cooperate rather than compete with AI. Knee-jerk reactions like the 3 January 2023 ban on ChatGPT from all school computers and Wi-Fi networks in New York City public schools will not work. This kind of reaction will cause more harm than good and demonstrates a general lack of knowledge or motivation to better the educational system and students' experiences with cutting-edge new technologies that are publicly available.

There are several open possibilities for future work of ChatGPT with AI. Some of them are:

- Adversarial attacks and defenses
- Multi-modal learning
- Explainable AI
- Human-AI collaboration
- Privacy-preserving AI

6.1 Key managerial insights

The adoption of AI-powered models like ChatGPT in cybersecurity can be a game-changer in the way that organizations manage the digitization risks they face, with strategic and operational benefits to decision-makers. In a proactive and automated protection sense, ChatGPT can also facilitate the task of discovering and thwarting cyber threats by automating the steady routine supervision, examining massive loads of log entries with unrivaled speed, and making quick suggestions of containment, giving managers time to devote their resources to additional priority strategic developments instead of manual detection. The capacity to

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Table 8 The distances between each FR_i (where I ranges from 1 to 10) and FPIDSol based on each NFReq_i (where I ranges from 1 to 10)

Freq and FPISSol (Distance)	NFReq1	NFReq2	NFReq3	NFReq4	NFReq5	NFReq6	NFReq7	NFReq8	NFReq9	NFReq10
d (Freq1, FPIDSol)	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78
d (Freq2, FPIDSol)	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75
d (Freq3, FPIDSol)	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78
d (Freq4, FPIDSol)	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78
d (Freq5, FPIDSol)	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75
d (Freq6, FPIDSol)	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78
d (Freq7, FPIDSol)	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78
d (Freq8, FPIDSol)	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75
d (Freq9, FPIDSol)	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78
d (Freq10, FPIDSol)	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78

Table 9 The distances between each FR_i (where I ranges from 1 to 10) and FNIDSol based on each NFReq_i (where I ranges from 1 to 10)

Freq and FNISSol (Distance)	NFReq1	NFReq2	NFReq3	NFReq4	NFReq5	NFReq6	NFReq7	NFReq8	NFReq9	NFReq10
d (Freq1, FNIDSol)	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.07
d (Freq2, FNIDSol)	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.07
d (Freq3, FNIDSol)	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08
d (Freq4, FNIDSol)	0.14	0.14	0.14	0.14	0.14	0.14	0.14	0.14	0.14	0.14
d (Freq5, FNIDSol)	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.07
d (Freq6, FNIDSol)	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08
d (Freq7, FNIDSol)	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08
d (Freq8, FNIDSol)	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.07
d (Freq9, FNIDSol)	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08
d (Freq10, FNIDSol)	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08

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Table 10 Distance between Ai and A* with respect to each criterion

Freq and (Ai, A*) (Distance)	NFReq1	NFReq2	NFReq3	NFReq4	NFReq5	NFReq6	NFReq7	NFReq8	NFReq9	NFReq10
d (Freq1, (Ai, A*))	0.42	0.51	0.51	0.43	0.37	0.53	0.45	0.36	0.38	0.36
d (Freq2, (Ai, A*))	0.36	0.34	0.55	0.41	0.55	0.42	0.49	0.34	0.36	0.39
d (Freq3, (Ai, A*))	0.48	0.35	0.48	0.41	0.57	0.43	0.49	0.34	0.35	0.36
d (Freq4, (Ai, A*))	0.37	0.36	0.36	0.42	0.42	0.57	0.43	0.35	0.40	0.56
d (Freq5, (Ai, A*))	0.41	0.36	0.35	0.41	0.51	0.41	0.48	0.34	0.40	0.46
d (Freq6, (Ai, A*))	0.35	0.34	0.34	0.51	0.48	0.46	0.46	0.34	0.41	0.52
d (Freq7, (Ai, A*))	0.41	0.34	0.34	0.47	0.41	0.50	0.53	0.34	0.34	0.50
d (Freq8, (Ai, A*))	0.36	0.34	0.36	0.46	0.46	0.55	0.46	0.34	0.36	0.46
d (Freq9, (Ai, A*))	0.36	0.34	0.34	0.36	0.47	0.51	0.34	0.34	0.41	0.50
d (Freq10, (Ai, A*))	0.46	0.36	0.35	0.56	0.53	0.51	0.47	0.34	0.35	0.50

Table 11 Distance between Ai and A- (Negative) with respect to each criterion

Freq and (Ai, A-) (Distance)	NFReq1	NFReq2	NFReq3	NFReq4	NFReq5	NFReq6	NFReq7	NFReq8	NFReq9	NFReq10
d (Freq1, (Ai, A-))	0.46	0.42	0.42	0.45	0.48	0.33	0.46	0.50	0.48	0.50
d (Freq2, (Ai, A-))	0.46	0.48	0.23	0.44	0.23	0.43	0.41	0.48	0.46	0.46
d (Freq3, (Ai, A-))	0.43	0.49	0.43	0.46	0.23	0.45	0.43	0.51	0.49	0.48
d (Freq4, (Ai, A-))	0.42	0.44	0.44	0.40	0.40	0.19	0.40	0.44	0.43	0.20
d (Freq5, (Ai, A-))	0.44	0.46	0.48	0.44	0.32	0.44	0.41	0.48	0.45	0.42
d (Freq6, (Ai, A-))	0.49	0.51	0.50	0.33	0.43	0.45	0.45	0.51	0.46	0.33
d (Freq7, (Ai, A-))	0.46	0.51	0.50	0.44	0.46	0.34	0.33	0.50	0.51	0.34
d (Freq8, (Ai, A-))	0.46	0.48	0.46	0.43	0.42	0.23	0.43	0.48	0.46	0.42
d (Freq9, (Ai, A-))	0.48	0.51	0.50	0.48	0.44	0.33	0.50	0.51	0.46	0.34
d (Freq10, (Ai, A-))	0.45	0.48	0.49	0.24	0.33	0.33	0.44	0.51	0.49	0.34

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Table 12 Closeness coefficients and ranking order of FReq's

FReq's	Closeness coefficients	Ranking order
$FReq_1$	0.53	4
$FReq_2$	0.57	2
FR_3eq	0.53	5
$FReq_4$	0.48	6
$FReq_5$	0.44	9
$FReq_6$	0.42	10
$FReq_7$	0.48	7
$FReq_8$	0.59	1
$FReq_9$	0.56	3
$FReq_{10}$	0.46	8

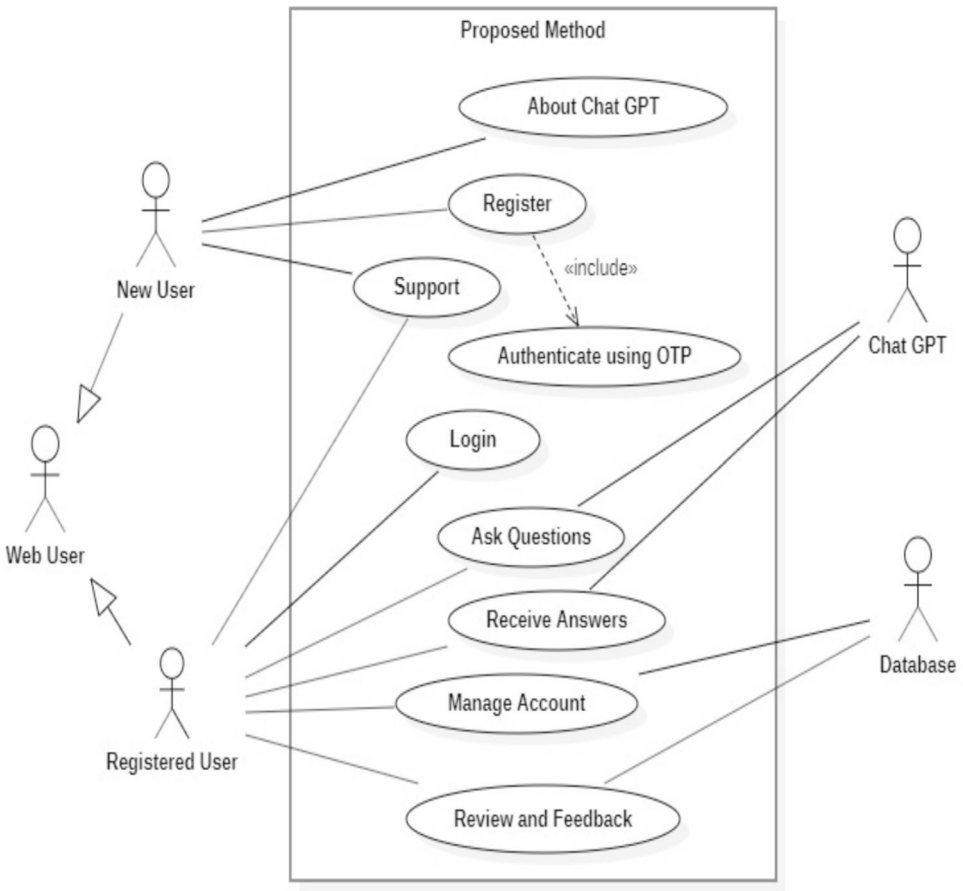


Fig. 4 UCD of the proposed approach

provide improved threat intelligence is owed to the capability to synthesize data collected from various structured and unstructured sources, such as security bulletins, open-source intelligence (OSINT) and internal reports of incidents into usable intelligence that can be utilized by the managers in implementing proper targeted defense in the area. This results in enhanced efficiency in security since the risks are prioritized, and depending on such findings, the managers will be able to decide upon applying the budgets, manpower, and technological investments in the future with more definite precision instead of following the general security regulations. By iteratively observing and adjusting, ChatGPT may assist in developing dynamic risk dashboards that would adapt to the latest actions and events in real time so that elected policies, controls, and responses could be changed on par with changes to the threatening environment. Through its ability to analyze, data-driven decision-making

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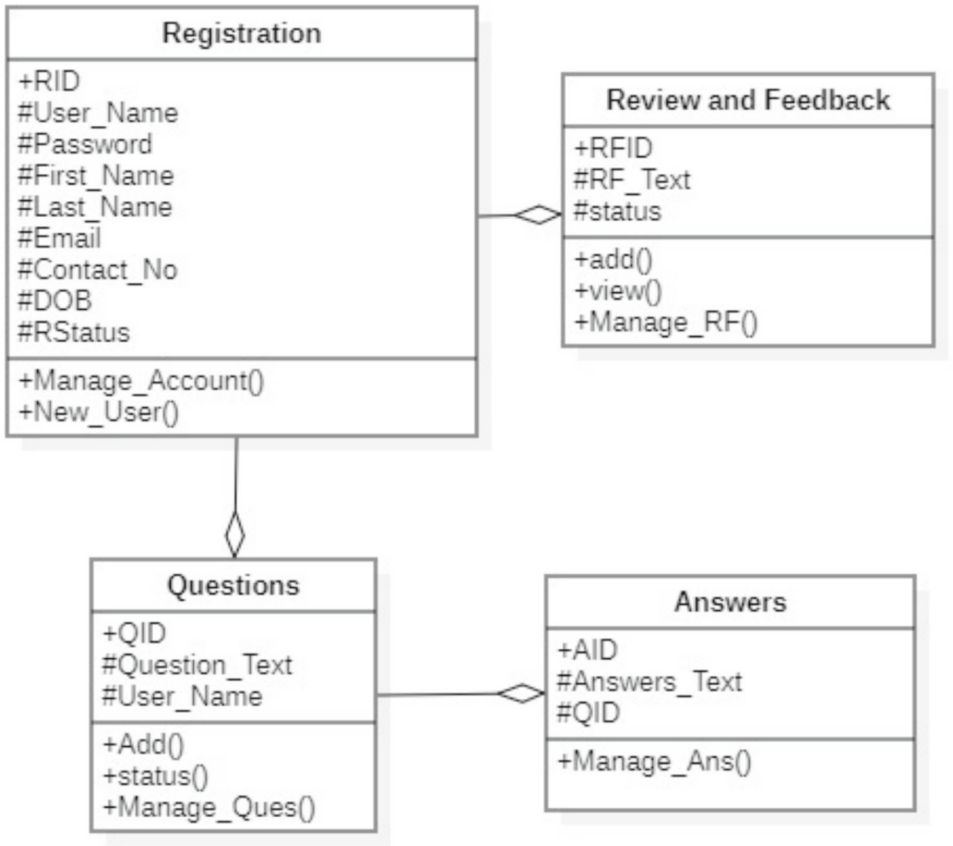


Fig. 5 Class diagram of the proposed approach

is also achievable, and leaders can quantitatively evaluate the security posture, measure ROI on security initiatives, and base their choices on evidence-based measurements. The implementation of intelligent systems of this kind is most importantly also increasing the emphasis on managerial decision-making, given that the leadership is enabled to make informed, timely, and strategic decisions on matters of cyber defense policies, incident response management, and long-term strategic resilience plans. Not only do these abilities decrease the risk of malicious access, but they also enhance the organizational responsiveness to some threats, their stakeholder trust, and the ability to address cybersecurity measures as part of business processes, proving the practical value and adaptiveness of the given research to the companies aiming at safeguarding their resources and remaining competitive in the digital age.

7 Conclusion and future directions

This paper introduces a developed methodology for modeling a specific set of FRs in a ChatGPT system. The method utilizes UCD principles and class diagrams to represent and analyze the FRs effectively.

This method utilizes fuzzy TOPSIS to compute the ranking values of FRs in a ChatGPT system, considering multiple DMs' opinions. A VC + + program was developed to facilitate the computation of ranking values to handle large requirements. The FRs and NFRs for the ChatGPT system were identified using a goal-oriented approach, and opinions from five DMs were collected to determine the ranking order of the FRs. The analysis revealed that various modeling techniques, such as UML class diagrams, UCDs, activity diagrams, SysML, and Tropos, are commonly used in the literature to represent ChatGPT requirements. UCDs and class diagrams were particularly useful for expressing system requirements and requirements analysis. This paper has explored the practices of AI-based cybersecurity and proposed strategies generated by ChatGPT to combat cybercrime. The paper has highlighted how technology can understand and analyze the human language used in cybercrime activities by leveraging ChatGPT's natural language processing capabilities. This approach enables ChatGPT to identify patterns and anomalies in data that traditional cybersecurity solutions might miss, allowing it to detect and respond to threats more promptly and effectively. Furthermore, the paper has emphasized integrating ChatGPT with other security tools to create a comprehensive and adaptive

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Table 13 Comparative Analysis of Language Models

Feature	ChatGPT	GPT-3 [79]	BERT [80]	Transformer-XL [81]	Bard
Number of Parameters	175 billion	175 billion	340 million—1.1 billion	257 million	137B parameters (as of 2022)
Training Data	Web text	Web text	Books, articles, and web pages	Web text	Google Search results, public web documents, and other Google datasets
Pre-training Method	Unsupervised	Unsupervised	Unsupervised	Unsupervised	Unsupervised with a focus on factual language understanding
Strengths	Strong conversational AI, good at summarization and question-answering	Strong conversational AI, good at various NLP tasks	Good at sentiment analysis and text classification	Handles long dependencies well	Strong factual language understanding, informative and comprehensive responses, ability to access and process real-time information (with Bard Pro)
Weaknesses	Can be prone to factual errors and biases	Limited compared to later models in some applications	Requires fine-tuning for specific tasks	High training cost	It may require further development for creative content generation capabilities
Applications	Conversational AI, language translation, summarization, question answering, chatbots	Like ChatGPT	Sentiment analysis, text classification, question answering, chatbots	Language modeling, text classification, and question answering	Like ChatGPT, with the addition of real-time information access and potential for research and code generation (limited access)

security system. Therefore, the paper recommends implementing ethical guidelines to ensure the responsible use of ChatGPT and other AI-based cybersecurity solutions. Moving forward, our future focus will be directed towards the following areas:

- The analysis of FRs and NFRs in the context of the ChatGPT system was conducted using a goal-oriented approach along with UML diagrams. The goal-oriented approach allowed for identifying and organizing the FRs and NFRs, while UML diagrams served as a visual representation to analyze and understand the requirements.
- A comparison was made between structural and behavioral UML diagrams when dealing with a large set of requirements in a ChatGPT system.
- As AI evolves, these solutions will likely become even more sophisticated, leading to a safer and more secure digital world for all.
- Future research will focus on applying the proposed framework to real-world ChatGPT-related cybersecurity datasets, such as incident logs or policy enforcement scenarios, to empirically validate its effectiveness.

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Table 14 Comparative Analysis of Requirement Prioritization: Fuzzy TOPSIS vs. AHP

Requirement	Fuzzy TOPSIS Rank	AHP Rank	Remarks
F_{Req8} (Knowledge Retrieval)	1st	2nd	Top-ranked in both; consistent importance
F_{Req10} (Personalization)	2nd	4th	Ranked higher in fuzzy due to linguistic nuance
F_{Req9} (Error Handling)	3rd	3rd	Identical ranking, high consensus
F_{Req5} (Intent Recognition)	4th	1st	AHP favors it more due to precise scoring
F_{Req1} (Natural Language Understanding)	5th	5th	Balanced ranking in both methods
F_{Req4} (Multi-turn Conversation)	6th	6th	Stable mid-level priority
F_{Req3} (Context Awareness)	7th	7th	Matching importance
F_{Req7} (Sentiment Analysis)	8th	8th	Low priority in both
F_{Req6} (Entity Recognition)	9th	9th	Lesser impact as per both methods
F_{Req2} (Natural Language Generation)	10th	10th	Consistently lowest in relevance

Appendix

Listofabbreviations.

AI	Artificial intelligence
SMEs	Small and medium-sized enterprises
UML	Unified modeling language
SRs	System requirements
TFNs	Triangular fuzzy numbers
FReq	Functional requirements
NFReq	Non-functional requirements
DMs	Decision makers
SH	Stakeholder
EC	Evaluate criteria
FDM	Fuzzy decision matrix
NFDM	Normalized fuzzy decision matrix
WNFDM	Weighted normalized fuzzy decision matrix
FPISol	Fuzzy positive idea solution
FNISol	Fuzzy negative idea solution
SG	Subgroup
H	High
L	Low
EH	Extremely high
EL	Extremely low
A	Average

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Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

Research involving human and /or animals Not applicable.

Informed consent Not applicable.

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